

5.3.11

AI25BTECH11017-BALU

Question:

Find whether the following pair of linear equations are consistent or inconsistent:

$$5x - 3y = 11, \quad -10x + 6y = 22 \quad (0.1)$$

Solution:

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

Given two lines

$$\mathbf{A} = \begin{pmatrix} 5 & -3 \\ -10 & 6 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 11 \\ 22 \end{pmatrix} \quad (0.2)$$

(0.3)

$$\mathbf{AX} = \mathbf{B} \quad (0.4)$$

rank of $\mathbf{A} = 1$

rank of $(\mathbf{A} \quad \mathbf{b}) = 2$

As both ranks are not equal the pair of linear equations are inconsistent

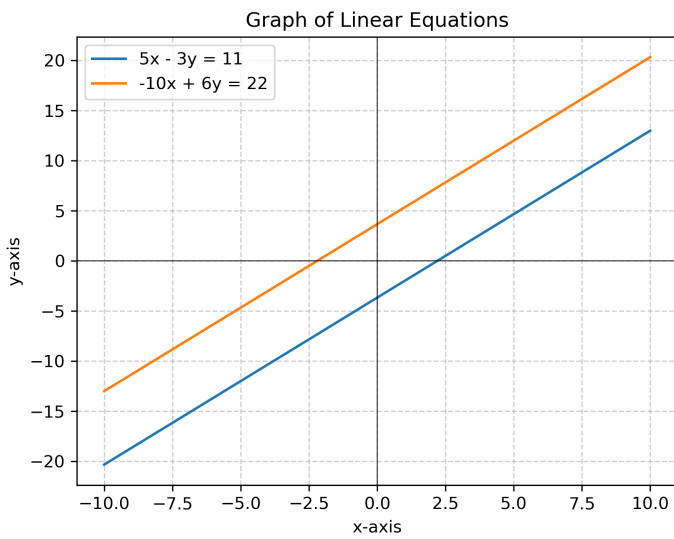


Fig. 0.1